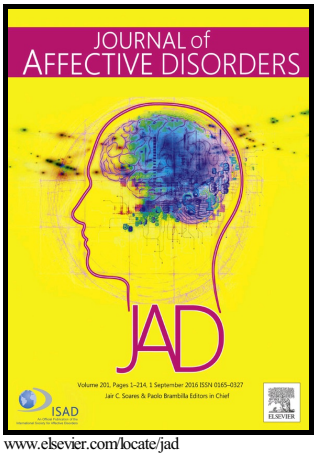
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Eye diagnosis differentiates seasonality outcomes in persons with severe visual impairment

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Eye diagnosis differentiates seasonality outcomes in persons with severe visual impairment

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Abstract

Background

Light plays a crucial role in both the pathogenesis and treatment of seasonal affective disorder (SAD). Consequently decreased retinal sensitivity to light has been suggested to be a risk factor for SAD. In a population of persons with severe visual impairment we recently found a highly increased prevalence of SAD. We now aimed to identify eye disorders or anatomical locations with specific association to seasonality.

Methods

In 912 cases (33%) from our prior seasonal pattern assessment questionnaire (SPAQ) screening study, we retrieved eye diagnoses from the Danish national register of patients and analyzed for specific eye disorders or anatomical locations that significantly differentiated SPAQ outcomes (global seasonality score, GSS and SPAQ-SAD prevalence).

Results

Persons with early life eye disorders (congenital conditions or retinopathy of prematurity) reported less symptoms of SAD (median GSS 4.5) than persons with acquired eye disorders (median GSS 5.0, p=0.005). Persons with macular degenerative disorders (MD) had highly increased seasonality outcomes (hazard ratio 2.23, p=0.002, median GSS 5 vs. 8, p=0.01).

Limitations

the study is a cross-sectional study based on a self-report questionnaire. Register data may be incomplete.

Conclusions

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MD is significantly associated to high-level seasonality and SAD prevalence. Early life eye disorder is associated to slightly lower seasonality compared to acquired eye disorder. Longitudinal studies are needed to assess causality.

Introduction

Seasonal mood swings are widespread in countries with large seasonal variation in photoperiod (Magnusson, 2000). The syndrome of seasonal affective disorder (SAD) was initially described in 1984 and today it appears as a subtype of major depressive disorder in the DSM-5 (American Psychiatric Association, 2013; Rosenthal et al., 1984). The criteria for SAD are depressive episodes with a recurrent seasonal pattern of onset and remission, absence of non-seasonal major depressive episodes for at least two years, and a high degree of recurrence over the lifespan. SAD is characterized by classical depressive symptoms as decreases in mood and energy levels and social withdrawal but is often accompanied by atypical symptoms of hypersomnia and hyperphagia, specifically a carbohydrate craving which often lead to weight gain (Rosenthal et al., 1984). SAD is highly prevalent in countries with large seasonal variation in photoperiod where prevalence rates as high as 8-12 % are reported in northern parts of the United States and Scandinavia when applying the frequently used screening tool, the Seasonal Pattern Assessment Questionnaire (SPAQ) (Magnusson, 2000). Studies using clinical interview or the more diagnostically valid Seasonal Health Questionnaire consistently report lower estimates with rates around 2-5 % for central European countries (Pjrek et al., 2016; Thompson et al., 2004).

Light treatment (first choice treatment), psychotherapy and antidepressant medication are effective treatments (Golden et al., 2005; Rohan et al., 2009, 2016) but the intertwining neurobiological, cognitive and environmental factors leading to the condition remain unclear. Light is suggested to play a key role in the pathogenesis of SAD due to the temporal and geographical association to photoperiodic variety and the strong treatment effect of light. It has been suggested that the retinal sensitivity to light is reduced in SAD patients who consequently are unable to maintain optimal light input to the brain when light levels are poor (Rohan et al., 2009). Clinical investigations attempting to address this hypothesis have produced inconsistent evidence for the hypothesis, which may be partly attributable to the diversity of the methodologies used. Electrooculographic ratios (EOG), which measure the

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total retinal electrical response to light, seem to be diminished in SAD patients whereas it has not been possible to find consistent evidence for any specific photoreceptor dysfunction. In recent years pupillometry has been introduced as a means to investigate the function of the small number of intrinsically photosensitive retinal ganglion cells (ipRGCs), which play a role in the unconscious non-image forming visual functions of light transmission. Although the blue light sensitive ipRGCs are few in number and only convey a fraction of the total light input, their function have been attempted linked to affective disorder because of their projections from the retina and to brain regions with influence on circadian function and mood regulation (Roecklein et al., 2013). Empirical evidence for this association is however rather sparse. One pupillometric study has found a reduced ipRGC function in SAD patients (Roecklein et al., 2013) and one study has reported an effect of blue-blocking glasses as add-on treatment for acute mania (Henriksen et al., 2016). It remains unclear whether the effect of light therapy for depression is attributable to blue or other wavelengths (Anderson et al., 2016).

In recent years there have been multiple reports of an increased risk of depression among persons with visual impairment but the condition is often overlooked and left untreated (Nollett et al., 2016). We recently reported the finding of an increased prevalence of SAD in this population (Madsen et al., 2016). In the present study we aimed to identify specific subgroups with increased prevalence of SAD symptoms. Apart from identifying potential vulnerable groups we believe such findings may contribute to the clarification of the pathogenesis of SAD.

Methods

Population

All members of the Danish Association of the Blind (DAB) between 18 and 65 years (2781 persons) were invited to our primary study investigating the prevalence of SAD symptoms among persons with severe visual impairment. Of these, 1647 persons (59%) completed the questionnaire. The DAB is a private organization of approximately 12.000 persons, which is assumed to be approximately 50% of all eligible Danish citizens with visual impairment or blindness. Membership is only offered to persons with residual vision below 6/60 Snellen visual acuity. To be as accepted as a member, residual vision has to be documented by an ophthalmologist.

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Survey

The study was performed in a cross-sectional design using audio, large print or Braille font questionnaires and telephone follow up (Madsen et al., 2012, 2016). Questionnaires were distributed in late October 2010. The letters contained the questionnaires, an information letter, a stamped return envelope and information that replies could be returned in email, post or to an open telephone line. Moreover the members were informed that they would be contacted by telephone and offered a telephone interview if no reply was received within 2 months. All replies were thus completed between late October and February.

Questionnaire

The Seasonal Pattern Assessment Questionnaire (SPAQ) measures seasonal variation (0-4 points) in 6 items, which are sleep duration, energy level, mood, body weight, appetite and social activity level. The items combine to a global seasonality score (GSS) with a range of 0-24. Moreover the SPAQ rates to which degree the seasonal variation is a problem and identifies the time of year of most discomfort. The SPAQ criteria for SAD (SPAQ-SAD) are 1) a GSS > 10, 2) seasonal variation considered a problem to at least a moderate degree and 3) winter time discomfort (November-February)(Kasper et al., 1989). In addition to completing the SPAQ, respondents reported their ability to perceive light, to determine direction of light or to count fingers at 1 meter.

Eye diagnoses

The National Danish Register of Patients contains information of all in- and outpatient contacts with Danish hospitals since 1977. Our register search included all diagnoses of eye disease or extensive eye surgery according to the applied ICD-8 and 10 classifications (Madsen et al., 2012). Eye disorders were consequently organized according to anatomical location (optical nerve, retina, vitreous humor, iris, lens, cornea and miscellaneous (congenital an- or microphthalmia, bulbar degeneration, degenerative myopia, endophthalmitis). The major disorders within the optical nerve and retina were tested specifically (glaucoma, optical atrophy, diabetic retinopathy, retinitis pigmentosa (RP), retinopathy of prematurity (ROP), retinal detachment and macular degeneration (MD)).

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We computed a group of diagnoses with perinatal onset "early life eye disorders" including all visually impairing conditions registered as congenital (in ICD-10 labelled "DQ") including congenital cataract, congenital glaucoma, congenital retinal degeneration, microphthalmia and ROP.

Analysis

Data were analyzed in SPSS version 22. Gross comparisons were made by use of Mann Whitney U, Chi2-tests or Fisher's exact test when appropriate. Eye conditions with statistically significant effects on seasonality parameters (GSS or SPAQ-SAD prevalence) ($p<0.05$) were tested in a general linear regression model and a logistic regression model adjusting for gender, age, reply method, light perception status (+/- light perception) and other major eye disorders (glaucoma, optical atrophy, diabetic retinopathy, RP, ROP, retinal detachment and MD). Only winter-type seasonality was included in the analysis why persons with non-seasonal or summer pattern were excluded from the analysis of GSS.

Ethics

The study was approved by the Danish Data Protection Agency. The study is an observational study and as such it does not need approval from any ethical committee according to Danish law.

Results:

Of the 1647 participants enrolled in the original survey, 1347 stated their eye diagnosis and/or gave permission for us to perform linkage to the registers. In 912 cases it was possible to retrieve one or more eye diagnoses from the register, which corresponds to 33 % of the 2781 persons invited to participate in the study. The majority were registered with more than one eye diagnosis (table 1), the two most frequent being glaucoma (23%) and cataract (56%). In cases of bilateral enucleation, prior eye diagnoses were not included. Respondents with summer (n=24) or a non-seasonal pattern (n=9) of discomfort were not included in analysis of eye disorder effect on GSS to ensure that we were testing for winter seasonality exclusively.

Persons registered with an early life eye disorder (n=193) had significantly lower GSS (median 4.5, p=0.005) than persons with eye disorders acquired later in life (median 5.0)

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(table 2). In multivariate analysis adjusting for gender, age, interview method, light perception status and other eye disorder this effect remained significant (p=0.01). For SPAQ-SAD prevalence the difference was not significant (17% versus 20%, p=0.36).

Persons registered with MD (n=90) had significantly increased GSS (median 8.0) and SPAQ-SAD prevalence (32.2%) compared to persons without a diagnosis of MD (GSS median 5.0, SPAQ-SAD 18.2%, p<0.01). This effect remained significant in multivariate analysis of GSS (p<0.01) adjusting for gender, age, interview method, light perception status and other eye disorder. The hazard ratio for SPAQ-SAD in MD was 2.23 (95% confidence intervals 1.35-6.74, p=0.002) in logistic regression analysis.

There was no significant association between seasonality parameters (GSS and SPAQ-SAD prevalence) and any anatomical location of eye disorder (optical nerve, retina, vitreous humor, iris, lens or cornea), nor between seasonality parameters and any specific eye diagnosis (glaucoma, optical atrophy, RP, ROP or retinal detachment) besides MD. For persons with diabetic retinopathy the prevalence of SAD was significantly increased from 18.1% to 25.0%, but there was no difference in GSS (p=0.65).

Twenty persons had bilateral anophthalmia. Of these, only one person had a GSS above 10. Although a highly interesting finding, the lack of power in this subgroup prevents statistical testing.

Discussion:

We recently reported a highly increased prevalence of SAD symptoms (SPAQ-SAD 17%) in a population with severe visual impairment or blindness (Madsen et al., 2016). Through subsequent linkage to the Danish Register of Patients we find that persons with early life eye disorder report less mood seasonality than persons with eye disorders acquired later in life. Persons with eye disorders causing degeneration of the macula are more prone to report symptoms of SAD than persons with other eye disorders. The causality of the association between visual impairment and mood seasonality is unknown but it is plausible that visual impairment limit light input to the brain with consequent effects on mood and energy levels. Some eye conditions physically block the light input through the eye whereas others impair the regulation of the retinal light sensitivity causing a suboptimal light input.

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Persons with early life eye disorder report lower seasonal variation in the measured mood and behavioral parameters. This finding is in line with other research addressing the impact of perinatal light conditions on neurobiological systems. Thus animal studies have shown that both retinal and circadian systems undergo persistent modifications depending on the perinatal light conditions (Ciarleglio et al., 2011; Jackson et al., 2014). Studies assessing the effect of season of birth on mood seasonality find that persons born during short photoperiods are less vulnerable to SAD than persons born during long photoperiods (Natale et al., 2007). It is possible that a perinatal lesion or blockage in ocular light transmission may lead to neurobiological alterations that mimic those seen in people born during short photoperiod and accordingly lead to a reduced vulnerability to develop SAD.

Age-related macular degeneration (AMD) is the primary cause of severe visual impairment and blindness in developed countries. Recent studies show that depression may be overrepresented in AMD compared to other eye disorders (Cimarolli et al., 2015) which might represent a bias to our finding. However, the SPAQ measures seasonal mood variation and not current depressive symptoms why we do not believe that our finding of increased SAD is attributable to an increased prevalence of depression. AMD causes degeneration of the outer retinal layers including the photoreceptors and the condition is therefore not expected to affect the ipRGCs. However recently a pupillometric study found a reduced ipRGC function in early stage AMD (Maynard et al., 2015). A reduced ipRGC function may predispose persons to develop SAD through reductions of blue light signals, possibly even more so in combination with the use of tinted eyewear. This is often advised in AMD since there is some evidence for sun exposure being a risk factor for AMD. However, as our data give no suggestion to the causality of the association between SAD and MD the reverse relationship is also plausible. Graw et al showed that SAD patients increase their outdoor light exposure in summer compared to healthy controls (Graw et al., 1999) which may increase their risk of retinal photo damage and in turn to AMD and visual impairment. These relations can only be investigated in longitudinal studies. If AMD is a risk factor for SAD, it is a highly prevalent one. It seems of relevance to screen for seasonal components of depressive symptoms when strategies to handle the increased risk of depression among visually impaired are implemented. Currently there are however only limited knowledge of effect and tolerability of

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light therapy in populations with severe eye disorder or visual impairment (Partonen et al., 1995).

The post-illumination pupillary response to blue light (PIPR) is currently the only measure of ipRGC function and has not yet been assessed across all eye disorders. In glaucoma there seem to be significant damage to ipRGCs (Cui et al., 2015) but our data showed no evidence for increased seasonality in this group or in other retinal diseases. The finding of an increased SPAQ-SAD prevalence among persons with diabetic retinopathy was not accompanied by a difference in GSS and we do therefore not consider it a valid expression of increased seasonality.

There are some limitations to consider. The cross-sectional design and the self-report of seasonality and light perception pose inherent risks of bias. The SPAQ is not a diagnostic tool but is acceptable for screening purposes. The National Danish Register of Patients does not include hospital contacts prior to 1977 or information of contacts to private ophthalmologists where the majority of eye disorders are handled. In 735 cases (26%) it was not possible to retrieve eye diagnoses. We expect these cases to have permanent impairments with onset prior to 1977, e.g. eye injury, but also progressive eye diseases without treatment potential. Of the non-responders, 9% were deceased or had severe competing illness or disability, 9% refused to participate and 23% could not be reached. The rather low response rate poses a risk of selection bias but the effect cannot be determined. Our data offer no means to identify a reliable onset of eye disorder or visual impairment.

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Conclusions

Persons with acquired eye disorders have increased mood seasonality compared to persons with early life eye disorders. Macular degenerative disorders are associated with increased seasonality. Longitudinal studies are needed to establish causality. When handling depression among persons with visual impairment, a potential seasonality pattern should be assessed as this may influence treatment strategy.

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Table 1. Prevalence of eye disorders

	N (%)
No eyes	21 (2)
Optical nerve	**291 (32)**
Glaucoma	211 (23)
Optical atrophy	86 (9)
Retina	**535 (59)**
Diabetic retinopathy	173 (19)
Retinitis pigmentosa	164 (18)
Retinopathy of prematurity	59 (9)
Retinal detachment	164 (18)
Macular degeneration	90 (10)
Vitreous humor or anterior segment (excluding cataract)	**89 (10)**
Cataract	**492(56)**
Miscellanous	**21 (2)**
Early life eye disorder [a]	**193 (2)**

[a]Congenital visually impairing conditions and retinopathy of prematurity

Table 2. Global seasonality score (GSS) and prevalence of SPAQ-SAD in early life eye disorders and macular degeneration

	n	GSS median	GSS mean	GSS standard deviation	SPAQ-SAD n	SPAQ-SAD prevalence
Early life eye disorder	193	4.5	5.1	4.2	33	17.1
No early life eye disorder	719	5.0	6.3	5.1	145	20.2
Significance		0.005*				0.36**
Adjusted significance		0.01[a]				0.15[b]
Macular degeneration	90	8.0	8.2	5.8	29	32.2
No macular degeneration	822	5.0	5.8	4.8	149	18.1
Significance		<0.001*				0.001**
Adjusted significance		<0.01[a]				0.002[b]

GSS: global seasonality score;SPAQ: seasonal pattern assessment questionnaire.
*Mann-Whitney U, ** Chi^2, 2-sided, [a]General linear model adjusted for gender, age, reply method and light perception, [b]Logistic regression adjusted for gender, age, reply method, light perception and other eye disorders (glaucoma, optical atrophy, diabetic retinopathy, retinitis pigmentosa, retinopathy of prematurity, retinal detachment, macular degeneration/early life eye disorder)

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Highlights

- Eye diagnoses differentiate seasonality in persons with severe visual impairment
- Early life eye disorder is associated with lower seasonality scores than acquired eye disorders.
- Macular degeneration is associated with high-level seasonality

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